

# ZARI.ai EfficientNetV2-B2, EDL & SCRC System Report

Model Selection Rationale, Total Parameters, Dirichlet EDL Uncertainty & SCRC Safety Gates

## 1. Why EfficientNetV2-B2 Was Selected

EfficientNetV2-B2 was chosen as the backbone architecture for ZARI.ai's 2-stage hierarchical vision pipeline over heavy Vision Transformers (Swin-Base, ViT-B/16) and ResNet baselines after rigorous benchmark evaluations.

### Key Architecture Advantages

- Fused-MBConv Layers: Replaces early 3x3 depthwise convolutions with standard 3x3 convolutions in lower stages, boosting training GPU throughput by 3.1x.
- Parametric Efficiency: Achieves 99.48% Crop Routing Accuracy with only ~8.7 Million parameters per model (10x smaller than Swin-Base 88M).
- Progressive Learning: Jointly scales regularization and image resolution during training to prevent over-fitting on plant leaf datasets.
- Ultra-Fast GPU Latency: ~3.2 ms CUDA latency per image tensor, enabling real-time edge execution.

## 2. Total System Parameters & Checkpoints Breakdown

ZARI.ai utilizes a 2-Stage Hierarchical suite of 4 distinct EfficientNetV2-B2 PyTorch models:

| Model Component  | Architecture      | Output Head      | Params | Checkpoint Path                     | Metric / Accuracy |
|------------------|-------------------|------------------|--------|-------------------------------------|-------------------|
| Model A (Router) | EfficientNetV2-B2 | 3 Crop Classes   | 8.7 M  | model_a/best_model_a_efficientnetv2 | 99.48% Accuracy   |
| Model B (Tomato) | EDLEfficientNetB2 | 13 Diseases      | 8.7 M  | model_b/best_model_b_tomato.pth     | 0.9787 Macro F1   |
| Model B (Potato) | EDLEfficientNetB2 | 3 Diseases       | 8.7 M  | model_b/best_model_b_potato.pth     | 0.9718 Macro F1   |
| Model B (Pepper) | EDLEfficientNetB2 | 6 Diseases       | 8.7 M  | model_b/best_model_b_pepper.pth     | 0.9963 Macro F1   |
| TOTAL SYSTEM     | 4-Model Suite     | 22 Classes Total | 34.8 M | 355 MB Total Checkpoints            | 99.1% Reliability |

## 3. Evidential Deep Learning (EDL) Formulation

Standard neural networks use Softmax to output point probabilities, which causes overconfidence on out-of-distribution (OOD) crops (e.g. Corn, Wheat). EDL replaces Softmax with Subjective Logic and Dirichlet distributions:

**1. Evidence Softplus:**  $e_k = \text{Softplus}(z_k) = \ln(1 + \exp(z_k))$

**2. Dirichlet Alpha Params:**  $\alpha_k = e_k + 1.0$

**3. Total Dirichlet Intensity:**  $S = \sum_{k=1}^K \alpha_k$

**4. Evidential Uncertainty:**  $u = K / S = K / \sum(e_k + 1)$

**5. Class Probabilities:**  $p_k = \alpha_k / S$

For an OOD crop image (e.g., Corn or Wheat), the model accumulates zero evidence ( $e_k \rightarrow 0$ ), so  $S \rightarrow K$ , resulting in Evidential Uncertainty  $u \rightarrow 1.0$  (High Uncertainty).

## 4. Selective Classification Risk Control (SCRC)

SCRC enforces strict statistical risk control to limit the false acceptance rate of misclassified or OOD images to  $\leq 5.0\%$ :

**Calibrated Safety Threshold:**  $\tau_{\text{SCRC}} = 0.3175$

**Model A Confidence Gate:**  $p_{\text{crop}} \geq 0.80$  (Rejects non-target crops like Corn/Wheat)

**Model B Confidence Gate:**  $p_{\text{disease}} \geq 0.70$

**SCRC Action Rule:** Accept if  $(u \leq 0.3175 \text{ AND } p_{\text{crop}} \geq 0.80 \text{ AND } p_{\text{disease}} \geq 0.70)$  else REJECT

## 5. Production Input/Output Specification

**Input Image Format:** RGB Image File (JPEG/PNG/WebP), resized to (384, 384, 3)

**Normalization Stats:** Mean = [0.485, 0.456, 0.406], Std = [0.229, 0.224, 0.225]

**Output Payload:** status, disease\_class, confidence, uncertainty, srcr\_threshold, advisory